

Maria Teleki

SOCIALLY ROBUST AI

PhD Candidate in Computer Science at Texas A&M University in College Sta., TX
Avilés–Johnson Fellow, Google DeepMind Student Researcher (Summer '26)

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Summary

I build methods, benchmarks, and audits that improve AI robustness in socially complex situations. My research spans Speech AI and Computational Social Science, across three layers of social context:

Is AI robust to how people naturally speak?
→ TACL '26 (Conditionally Accepted), ICASSP '26, Interspeech '25, Interspeech '24, LREC-COLING '24

Is AI robust to each person's uniqueness?
→ ICWSM '25, Interspeech '26, ACL '26

Is AI robust within the sociopolitical structures it operates in?
→ FAccT '26

Education

2022 – Present | PhD Computer Science at Texas A&M University
Advised by James Caverlee. Defending May 4, 2027.

2017 – 2022 | B.S. Computer Science at Texas A&M University
Summa Cum Laude. Emphasis in Electrical Engineering. Minor in Mathematics.

Publications

P1. **The Role of Disfluencies in Speech Translation**
Maria Teleki*, Maike Züfle*, Vilém Zouhar, Fabian Retkowski, Oliver Grabner, Alex Waibel, Jan Niehues, and James Caverlee
— Collab. with KIT, ETH Zürich, Carnegie Mellon University
TACL '26 (Conditionally Accepted) Transactions of the Association for Computational Linguistics [Link]

P2. **The Due Process Deficit: Auditing AI Governance in U.S. Higher Education**
Maria Teleki, Anna Seo Gyeong Choi, Anne Duray, Haoran Liu, Julie Zhang, Xiangjue Dong, Dilma Da Silva, Allison Koenecke, and James Caverlee
— Collab. with Cornell, AAAS Science and Technology Policy Fellow
FAccT '26 The ACM Conference on Fairness, Accountability, and Transparency [Link]
→ *Presented at: IC2S2 Parallel Talk '26* and *The Contentious Politics of AI Network* (Remote)

P3. **Z-Scores: A Metric for Linguistically Assessing Disfluency Removal**
Maria Teleki, Sai Janjur, Haoran Liu, Oliver Grabner, Ketan Verma, Thomas Docog, Xiangjue Dong, Lingfeng Shi, Cong Wang, Stephanie Birkelbach, Jason Kim, Yin Zhang, and James Caverlee
ICASSP '26 IEEE International Conference on Acoustics, Speech and Signal Processing [Link]

P4. **The Voice Behind the Words: Quantifying Intersectional Bias in SpeechLLMs**
Shree Harsha Bokkahalli Satish, Christoph Minixhofer, ***Maria Teleki***, James Caverlee, Ondrej Klejch, Peter Bell, Gustav Eje Henter, and Éva Székely
— Collab. with KTH, University of Edinburgh
Interspeech '26 Conference of the International Speech Communication Association [Link]

P5. **CHOIR: Collaborative Harmonization fOr Inference Robustness**
Xiangjue Dong, Cong Wang, ***Maria Teleki***, Millennium Bismay, Ruihong Huang, and James Caverlee
ACL '26 *The Annual Meeting of the Association for Computational Linguistics* [Link]

P6. **I want a horror – comedy – movie: Slips-of-the-Tongue Impact Conversational Recommender System Performance**
Maria Teleki, Lingfeng Shi, Chengkai Liu, and James Caverlee
Interspeech '25 *Conference of the International Speech Communication Association* [Link]

P7. **Masculine Defaults via Gendered Discourse in Podcasts and Large Language Models**
Maria Teleki, Xiangjue Dong, Haoran Liu, and James Caverlee
ICWSM '25 *The International AAAI Conference on Web and Social Media* [Link]
→ *Presented at: IC2S2 Parallel Talk '25 and SICon@ACL '25*
→ *Media Coverage: MASKulinity Podcast*

P8. **A Survey on LLMs for Story Generation**
Maria Teleki, Vedangi Bengali, Xiangjue Dong, Sai Tejas Janjur, Haoran Liu, Tian Liu, Cong Wang, Ting Liu, Yin Zhang, Frank Shipman, and James Caverlee
EMNLP Findings '25 *Findings of the Association for Computational Linguistics: EMNLP 2025* [Link]
→ *Extended version: TIST '26 (Major Revision Submitted) [Link]*

P9. **Comparing ASR Systems in the Context of Speech Disfluencies**
Maria Teleki, Xiangjue Dong, Soohwan Kim, and James Caverlee
Interspeech '24 *Conference of the International Speech Communication Association* [Link]

P10. **Quantifying the Impact of Disfluency on Spoken Content Summarization**
Maria Teleki, Xiangjue Dong, and James Caverlee
LREC-COLING '24 *The Joint International Conference on Computational Linguistics, Language Resources and Evaluation* [Link]

P11. **Co2PT: Mitigating Bias in Pre-trained Language Models through Counterfactual Contrastive Prompt Tuning**
Xiangjue Dong, Ziwei Zhu, Zhuoer Wang, ***Maria Teleki***, and James Caverlee
— Collab. with George Mason University
EMNLP Findings '23 *Findings of the Association for Computational Linguistics: EMNLP 2023* [Link]

Preprints

R1. **A Cross-Community Agenda for Speech AI**
Maria Teleki*, Kimi V. Wenzel*, Anna Seo Gyeong Choi*, Tobias Weinberg*, Shree Harsha Bokkahalli Satish*, Stephanny Sanchez*, Belu Ticona*, Ariadna Sanchez*, Yash Sonkar*, Aarti Mathur*, Christoph Minixhofer*, Abraham Glasser*, Raja Kushalnagar*, James Caverlee*, Minha Lee*, Shaomei Wu*, Alyssa Hillary Zisk*, Éva Székely*, Dylan Gaines*, Angelika Seeschaaf Veres*, Seray Ibrahim*, Nicholas Cummins*, and Allison Koenecke*
Under Submission at ARR *ACL Rolling Review* [Link]

R2. **SpeechSpectrum: A Framework for Speech-to-Text Representation Along the Linguistic Fidelity Spectrum**
Maria Teleki*, Anna Seo Gyeong Choi*, Miguel del Rio, James Caverlee, Corey Miller, and Allison Koenecke
— Collab. with Cornell, Rev AI
In Prep. for TOCHI *The ACM Transactions on Computer-Human Interaction* [Link]
→ *Presented at: IC2S2 Poster '26*

R3. **Beyond Single Ground Truth: Reference Monism as Epistemic Injustice in ASR Evaluation**
Anna Seo Gyeong Choi, ***Maria Teleki***, James Caverlee, Miguel del Rio, Corey Miller, and Hoon Choi
— Collab. with Cornell, Rev AI, Kangwon National University
In Prep. for ARR *ACL Rolling Review* [Link]
→ *Presented at: IC2S2 Poster '26*

R4. **Conversational Speech Reveals Structural Robustness Failures in SpeechLLM Backbones**
Maria Teleki, Sai Janjur, Haoran Liu, Oliver Grabner, Ketan Verma, Thomas Docog, Xiangjue Dong, Lingfeng Shi, Cong Wang, Stephanie Birkelbach, Jason Kim, Yin Zhang, Éva Székely, and James Caverlee
— Collab. with KTH
In Prep. for ARR *The ACL Rolling Review* [Link]

In Progress

G1. **"Well, actually–": Socially Reasoning About Interruptions in Conversation**
Maria Teleki, Rajat Hebbar, Brian Eoff, and James Caverlee
— Collab. with Google DeepMind
In Prep. for TOCHI *ACM Transactions on Computer-Human Interaction*

Posters, Demos, Technical Reports, & Mentee Papers

M1. **Walk a Mile in My Voice: Voice Conversion Shapes Trust, Attribution, and Empathy in Human–AI Speech Interactions**
Shree Harsha Bokkahalli Satish, ***Maria Teleki***, Christoph Minixhofer, Ondrej Klejch, Peter Bell, and Éva Székely
— Collab. with KTH, University of Edinburgh
IUI '26 (Poster) *The ACM Conference on Intelligent User Interfaces* [Link]
→ *Presented at: CHI '26 (Poster)*

M2. UG 1ST AUTHOR
PromptHelper: A Prompt Recommender System for Encouraging Creativity in AI Chatbot Interactions
Jason Kim, ***Maria Teleki***, and James Caverlee
CHI '26 (Poster) *The ACM CHI Conference on Human Factors in Computing Systems* [Link]
→ *Presented at: IC2S2 Poster '26*

M3. UG 1ST AUTHOR
SocialPulse: An Open-Source Subreddit Sensemaking Toolkit
Stephanie Birkelbach, ***Maria Teleki***, Nehul Bhatnagar, Peter Carragher, Xiangjue Dong, and James Caverlee
— Collab. with Carnegie Mellon University, Revionics
SocialLLM @ ICWSM '26 *The Workshop on Large Language Models for Social Reasoning and Simulation at the International AAAI Conference on Web and Social Media* [Link]
→ *Presented at: IC2S2 Parallel Talk '26*

M4. UG 1ST AUTHOR
PodChecker: An Interpretable Fact-Checking Companion for Podcasts
Anna Irmetova, Haoran Liu, ***Maria Teleki***, Peter Carragher, and James Caverlee
— Collab. with Carnegie Mellon University
MisD @ ICWSM '26 *The Workshop on Misinformation Detection in the Era of LLMs at the International AAAI Conference on Web and Social Media* [Link]

M5. UG 1ST AUTHOR
Detecting and Mitigating Demographic Bias in LLM-Based Resume Evaluation
Oluwadayo Bamgbelu, ***Maria Teleki****, Xiangjue Dong*, and James Caverlee
In Preparation for ARR (Short) *ACL Rolling Review* [Link]
→ *Presented at: IC2S2 Poster '26*

M6. MS 1ST AUTHOR
When Behavioral Traces Mislead: Evaluating Evidence Adequacy in Learning Analytics
Junhyeon Kwon, Jobin Varughese, ***Maria Teleki***, and Mohsen Dorodchi
— Collab. with Texas A&M University Center for Teaching Excellence & Office of the Provost, UNC Charlotte
In Prep. for LAK (Short) *The International Learning Analytics & Knowledge Conference* [Link]
→ *Presented at: IC2S2 Parallel Talk '26*

M7. **AI in the Loop: A Multimodal Assistant for Real-Time Lecture Comprehension**
Amanda Lacy, ***Maria Teleki***, Esau Hutcherson, Jobin Varughese, Jun Kwon, Frank Shipman, and Tracy Hammond
— Collab. with Texas A&M University Center for Teaching Excellence & Office of the Provost
Conference on Learning @ Scale (L@S '26)

M8. **A Survey on LLM Inference-Time Self-Improvement**
Xiangjue Dong*, ***Maria Teleki****, and James Caverlee
arXiv [Link]

M9. MS 1ST AUTHOR
DACL: Disfluency Augmented Curriculum Learning for Fluent Text Generation
Rohan Chaudhury, ***Maria Teleki***, Xiangjue Dong, and James Caverlee
LREC-COLING '24 *The Joint International Conference on Computational Linguistics, Language Resources and Evaluation* [Link]

M10. **Howdy Y'all: An Alexa TaskBot**
Majid Alfifi, Xiangjue Dong, Timo Feldman, Allen Lin, Karthic Madanagopal, Aditya Pethe, ***Maria Teleki***, Zhuoer Wang, Ziwei Zhu, and James Caverlee
Alexa Prize TaskBot Challenge Proceedings 2022 *Alexa Prize TaskBot Challenge* [Link]

Experience

Google DeepMind | Student Researcher | Cambridge, MA | June – Sep '26
See G1.

RetailMeNot | Software Engineering Intern | Austin, TX | May – August '21
BERT embeddings and PCA to surface which semantic dimensions of coupon content predicted CTR.

The Hi, How Are You Project | Volunteer | Austin, TX | May – Dec '20
Built the "Friendly Frog" Alexa Skill with the organization at the beginning of the COVID-19 pandemic to promote mental health.

RetailMeNot | Software Engineering Intern | Austin, TX | May – August '20
Built the "RetailMeNot DealFinder" Alexa Skill to help users activate cash back offers.

Silicon Labs | Applications Engineering Intern | Austin, TX | May – August '19
Built a Python library adopted by customers for real-time hardware bus traffic analysis across three protocols.

Service

Workshop Organization
Workshop Co-Organizer: **Speech AI for All: The What, How, and Who of Measurement**
Kimi V. Wenzel, Alisha Pradhan, ***Maria Teleki***, Tobias Weinberg, Robin Netzorg, Alyssa Hillary Zisk, Anna Seo Gyeong Choi, Jingjin Li, Raja Kushalnagar, Colin Lea, Abraham Glasser, Christian Vogler, Nan Bernstein Ratner, Ly Xīnzhèn M. Zhǎngsūn Brown, Allison Koenecke, Karen Nakamura, and Shaomei Wu
CHI '26 Workshop *The ACM CHI Conference on Human Factors in Computing Systems* [Link]

Program Committee Service
Program Committee (Reviewer) for Interspeech: '26
Program Committee (Reviewer) for IC2S2: '26
Program Committee (Reviewer) for ACL ARR: Aug '24, Oct '24, Dec '24, May '25, Oct '25, Jan '26
Program Committee (Reviewer) for ICWSM: Jan '24, May '24, Sep '24, Jan '25, Sep '25, Jan '26
External Program Committee (External Reviewer) for RecSys: '24

Teaching

Certifications
2025 | Professional Development Mastery Certificate in Instruction & Assessment | GRAD Aggies, CIRTL@TAMU
Completed multi-semester program in evidence-based teaching, including syllabus design & learning assessment.

Course Development
Spring 2026 | Course Developer | Texas A&M University, CSCE 676 Data Mining (Graduate Level)
Developed assignments that required students to use coding agents to test their implementations and make video-recorded technical interviews that trained students to state their assumptions on vague questions.

Spring 2026 | Course Developer | Texas A&M University, CSCE 289 GenAI for Engineers (Undergraduate Level)
Helped shape the LLM and lightweight-coding curriculum for AI-driven engineering. Built the progression from foundational concepts through hands-on API work to critical evaluation, so students leave able to use these tools competently and skeptically.

Teaching Assistantships
Spring 2026 | Teaching Assistant | Texas A&M University, CSCE 676 Data Mining & Analysis

Guest Lectures
Fall 2025 | Speech Language Models | Texas A&M University, CSCE 676 Data Mining & Analysis
Spring 2025 | IR Evaluation and Learning to Rank | Texas A&M University, CSCE 670 Information Storage & Retrieval

Earlier Teaching
Dec 2018 – Dec 2019 | Peer Teacher | Texas A&M University
Taught weekly review sessions and assisted TA at lab sessions for CSCE 111 and 121.
Sep 2016 – July 2017 | Afterschool Instructor | The Y (YMCA)
Taught weekly classes at local elementary schools and developed STEM curriculum: authored robotics & science instruction manuals for the Williamson County YMCA Afterschool program.
May 2016 – Aug 2016 | Camp Counselor | Camp Twin Lakes, The Y (YMCA)
Worked as a kindergarten camp counselor, and taught robotics specialty class in the afternoon for K–8th grade students.

Mentoring

★ *published co-author*; ♠ *preprint co-author*; ▲ *completed thesis*; ♦ *for course credit*; ♣ *no prior publications.* NLP = Texas NLP Symposium; SRW = Student Research Week.

MS Students

Rohan Chaudhury [★▲] *M.S. CS* – Led [M9]. Now at Amazon.

Sai Tejas Janjur [★♠♣] *M.S. CS* – [P3], [P8], [R4]. 🏆 1st Place, SRW. Now at NVIDIA.

Cong Wang [★♠♣] *M.S. Statistics* – [P3], [P5], [P8], [R4]. Presented at NLP.

Junhyeon Kwon [★♠♣] *M.S. CS* – Led [M6]; [M7].

Undergraduate Students

Soohwan Kim [★♦♣] *B.S. CS* – [P9]. Now at UPS.

Oliver Grabner [★♠♣] *B.S. CS* – [P1], [P3], [R4]. Presented at NLP & SRW. Intern at Samsung, Google.

Thomas Docog [★♠♣] *B.S. CS* – [P3], [R4]. Presented at NLP. Intern at Ramirez.

Ketan Verma [★♠♣] *B.S. CS* – [P3], [R4]. Intern at Samsung. M.S. at Cornell Tech.

Stephanie Birkelbach [★♠♦♣] *B.S. CS + Statistics* – Led [M3]; [P3], [R4]. REU at RIT w/ Ashique KhudaBukhsh.

Jason Kim [★♠♣] *B.S. CS* – Led [M2]; [P3], [R4]. Presented at NLP. 🏆 1st Place, SRW.

Anna Irmetova [★♦♣] *B.S. CS* – Led [M4]. Presented at SRW. Intern at Target.

Oluwadayo (Dayo) Bamgbelu [♣] *B.S. CS* – Led [M5]. Presented at NLP. Intern at Meta.

Awards

2022–2026 | Dr. Dionel Avilés '53 and Dr. James Johnson '67 Fellowship in Computer Science and Engineering
Multi-year fellowship supporting doctoral research in Computer Science.

2024, 2025, 2026 | Department of Computer Science & Engineering Travel Grant

2024 | CRA-WP Grad Cohort for Women
Selected participant; supported with travel support.

2017–2021 | President's Endowed Scholarship
Four-year scholarship supporting undergraduate studies.

2018 | Bertha & Samuel Martin Scholarship

Invited Talks & Media

Fall 2026 (Upcoming) | *Socially Robust AI* | King's College London | Remote

Fall 2026 | *Socially Robust AI* | Cornell Tech | Remote

Fall 2026 | *Socially Robust AI* | George Mason University | Remote

Fall 2026 | *The Due Process Deficit* | The Contentious Politics of AI | Remote

Fall 2026 | *10 Ways You Can Use AI Right Now* | JLL | Houston, Texas

Spring 2026 | *Masculine Defaults (Podcast Interview)* | MASKulinity Podcast | Remote

Spring 2026 | *Accountable AI* | IPT Global | Houston, Texas

Fall 2025 | *Conversational AI* | Revionics | Austin, Texas

Spring 2025 | *How does ChatGPT work? & My research!* | Texas A&M University, Club of Aggie Female Engineers (C.A.F.E.) | College Station, Texas

Fall 2024 | *The Other AI: An Intuitive Understanding of Artificial Intelligence* | Texas Tech University, School of Veterinary Medicine VBMA | Amarillo, Texas

References

James Caverlee
Professor, Department of Computer Science & Engineering at Texas A&M University
Tenneco Professor, College of Engineering at Texas A&M University
caverlee@tamu.edu

Dilma Da Silva
Regents Professor, Department of Computer Science & Engineering at Texas A&M University
dilma@tamu.edu

Allison Koenecke
Assistant Professor, Department of Information Science at Cornell Tech
Field Faculty Member, Department of Computer Science at Cornell Tech
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Éva Székely
Assistant Professor, Division of Speech, Music and Hearing at KTH Royal Institute of Technology, Sweden
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